

Linux for DevOps

Lesson 1: Linux Fundamentals

Linux Fundamentals - Study Notes

Key Concepts

1. **Operating System (OS) Structure** – How the OS manages hardware, software, and user interactions.
2. **Kernel vs. User Space** – The core component that controls hardware (kernel) and the environment where applications run (user space).
3. **Linux Distributions** – Pre built collections of the kernel, core utilities, and package managers (e.g., Ubuntu, Fedora, Arch).
4. **Filesystem Hierarchy** – Standard directory layout (`/`, `/home`, `/etc`, `/var`, etc.) and how files are organized.
5. **Basic Shell Commands** – Essential navigation and file manipulation commands (`ls`, `cd`, `pwd`, `mkdir`, `rm`).

Summary

Linux is an open source operating system that separates the core kernel from user space utilities, allowing flexibility and modularity. The kernel handles low level tasks such as process scheduling, memory management, and device drivers, while user space hosts applications and tools. Distributions bundle the kernel with a set of tools, libraries, and a package manager, creating a ready to use environment. Understanding the filesystem hierarchy and mastering a handful of shell commands gives you the foundation to explore more advanced topics like scripting, networking, and system administration.

Important Points

- **OS vs. Kernel**: The OS is the entire system; the kernel is the heart that talks directly to hardware.
- **Distributions**: Choose a distro based on your goals (e.g., Ubuntu for beginners, Arch for learning the internals).
- **Filesystem Hierarchy Standard (FHS)**: `/` is the root; `/home` holds user data; `/etc` contains configuration; `/var` stores variable data.
- **Shell Basics**:
 - `pwd` – print working directory.
 - `ls` – list directory contents (options: `-l`, `-a`, `-h`).
 - `cd` – change directory (use `cd ..` to go up, `cd ~` to go to home).
- **Permissions**: Files have read (`r`), write (`w`), and execute (`x`) bits for owner, group, and others.
- **Package Managers**:
 - Debian/Ubuntu: `apt`.

- Red Hat/Fedora: ``dnf`/`yum``.
- Arch: ``pacman``.

Examples

1. Navigating the Filesystem

```
```bash
```

#### Show current directory

```
pwd
```

**Output: /home/alice**

#### List all files, including hidden ones, in long format

```
ls -la
```

#### Output:

```
drwxr-xr-x 2 alice alice 4096 Apr 10 10:15 .
```

```
drwxr-xr-x 3 root root 4096 Apr 10 09:00 ..
```

```
-rw-r--r-- 1 alice alice 45 Apr 10 10:12 notes.txt
```

---

\*Explanation\*: ``pwd`` confirms you're in ``/home/alice``. ``ls -la`` lists every file, showing hidden ones (``.`` and ``.``) and detailed permissions.

### 2. Installing a Package (Ubuntu)

```
```bash
```

Update package list

```
sudo apt update
```

Install the ``htop`` monitoring tool

```
sudo apt install htop
```

Explanation: ``sudo`` runs the command with administrative rights. ``apt update`` refreshes the repository index, and ``apt install`` downloads and installs ``htop``.

Quick Review

1. **What is the kernel's primary role in an OS?**
2. **Name two common Linux distributions and one unique feature of each.**
3. **Explain the difference between `cd ..` and `cd ~`.**
4. **What does the `-l` option do when used with `ls`?**
5. **How do you install a package on a Debian based system?**

Further Reading

- **Linux Filesystem Hierarchy Standard (FHS)** – deep dive into directory roles.
- **Shell Scripting Basics** – automating tasks with Bash.
- **Package Management** – comparing `apt`, `dnf`, `pacman`, and `yum`.
- **Permissions and Ownership** – `chmod`, `chown`, `chgrp`.
- **Process Management** – `ps`, `top`, `kill`, `nice`.

These topics will build on the fundamentals and prepare you for more advanced system administration and development work. Happy learning!

